

friction cone; and if a contact is sliding, the force is on the edge of the friction cone.

General references on contact modeling and manipulation include Handbook of Robotics chapters [66, 142] and the texts by Mason [109] and Murray et al. [122].

12.6 Exercises

Exercise 12.1 Prove that the impenetrability constraint (12.4) is equivalent to the constraint (12.7).

Exercise 12.2 Representing planar twists as centers of rotation.

- (a) Consider the two planar twists $\mathcal{V}_1 = (\omega_{z1}, v_{x1}, v_{y1}) = (1, 2, 0)$ and $\mathcal{V}_2 = (\omega_{z2}, v_{x2}, v_{y2}) = (1, 0, -1)$. Draw the corresponding CoRs in a planar coordinate frame, and illustrate $\text{pos}(\{\mathcal{V}_1, \mathcal{V}_2\})$ as CoRs.
- (b) Draw the positive span of $\mathcal{V}_1 = (\omega_{z1}, v_{x1}, v_{y1}) = (1, 2, 0)$ and $\mathcal{V}_2 = (\omega_{z2}, v_{x2}, v_{y2}) = (-1, 0, -1)$ as CoRs.

Exercise 12.3 A rigid body is contacted at $p = (1, 2, 3)$ with a contact normal into the body $\hat{n} = (0, 1, 0)$. Write the constraint on the body's twist $\mathcal{V}$ due to this contact.

Exercise 12.4 A space frame $\{s\}$ is defined at a contact between a stationary constraint and a body. The contact normal, into the body, is along the $\hat{z}$ -axis of the $\{s\}$ frame.

- (a) Write down the constraint on the body's twist $\mathcal{V}$ if the contact is a frictionless point contact.
- (b) Write down the constraints on $\mathcal{V}$ if the contact is a point contact with friction.
- (c) Write down the constraints on $\mathcal{V}$ if the contact is a soft contact.

Exercise 12.5 Figure 12.29 shows five stationary “fingers” contacting an object. The object is in first-order form closure and therefore force closure. If we take away one finger, the object may still be in form closure. For which subsets of four fingers is the object still in form closure? Prove your answers using graphical methods.

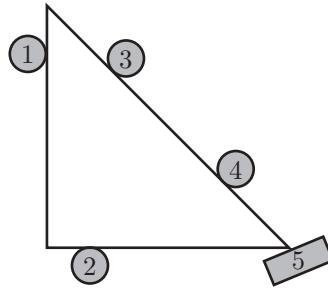


Figure 12.29: A triangle in contact with five stationary fingers, yielding first-order form closure and therefore force closure. Analyze the contact when one or more fingers are removed. The hypotenuse of the triangle is 45° from the vertical on the page, and contact normal 5 is 22.5° from the vertical.

Exercise 12.6 Draw the set of feasible twists as CoRs when the triangle of Figure 12.29 is contacted only by finger 1. Label the feasible CoRs with their contact labels.

Exercise 12.7 Draw the set of feasible twists as CoRs when the triangle of Figure 12.29 is contacted only by fingers 1 and 2. Label the feasible CoRs with their contact labels.

Exercise 12.8 Draw the set of feasible twists as CoRs when the triangle of Figure 12.29 is contacted only by fingers 2 and 3. Label the feasible CoRs with their contact labels.

Exercise 12.9 Draw the set of feasible twists as CoRs when the triangle of Figure 12.29 is contacted only by fingers 1 and 5. Label the feasible CoRs with their contact labels.

Exercise 12.10 Draw the set of feasible twists as CoRs when the triangle of Figure 12.29 is contacted only by fingers 1, 2, and 3.

Exercise 12.11 Draw the set of feasible twists as CoRs when the triangle of

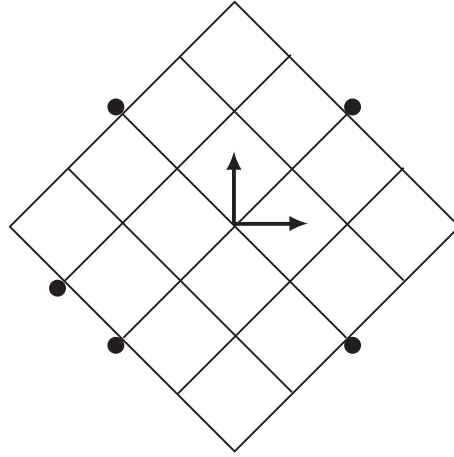


Figure 12.30: A 4×4 planar square restrained by five frictionless point contacts.

Figure 12.29 is contacted only by fingers 1, 2, and 4.

Exercise 12.12 Draw the set of feasible twists as CoRs when the triangle of Figure 12.29 is contacted only by fingers 1, 3, and 5.

Exercise 12.13 Refer again to the triangle of Figure 12.29.

- Draw the wrench cone from contact 5, assuming a friction angle $\alpha = 22.5^\circ$ (a friction coefficient $\mu = 0.41$), using moment labeling.
- Add contact 2 to the moment-labeling drawing. The friction coefficient at contact 2 is $\mu = 1$.

Exercise 12.14 Refer again to the triangle of Figure 12.29. Draw the moment-labeling region corresponding to contact 1 with $\mu = 1$ and contact 4 with $\mu = 0$.

Exercise 12.15 The planar grasp of Figure 12.30 consists of five frictionless point contacts. The square's size is 4×4 .

- Show that this grasp does not yield force closure.
- The grasp of part (a) can be modified to yield force closure by adding one frictionless point contact. Draw all the possible locations for this contact.

Exercise 12.16 Assume the contacts shown in Figure 12.31 are frictionless point contacts. Determine whether the grasp yields force closure. If it does not,

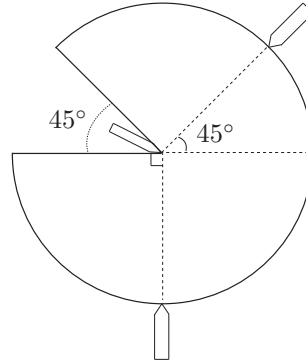


Figure 12.31: A planar disk restrained by three frictionless point contacts.

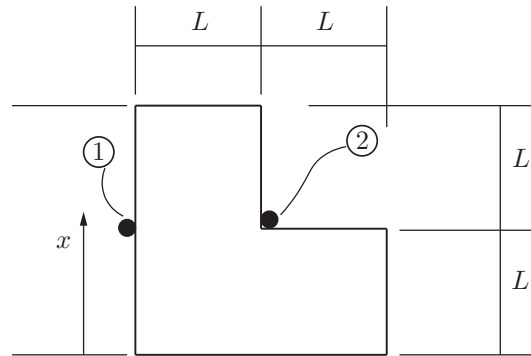


Figure 12.32: An L-shaped planar object restrained by two point contacts with friction.

how many additional frictionless point contacts are needed to construct a force closure grasp?

Exercise 12.17 Consider the L-shaped planar object of Figure 12.32.

- Suppose that both contacts are point contacts with friction coefficient $\mu = 1$. Determine whether this grasp yields force closure.
- Now suppose that point contact 1 has friction coefficient $\mu = 1$, while point contact 2 is frictionless. Determine whether this grasp yields force closure.
- The vertical position of contact 1 is allowed to vary; denote its height by x . Find all positions x such that the grasp is force closure with $\mu = 1$ for

contact 1 and $\mu = 0$ for contact 2.

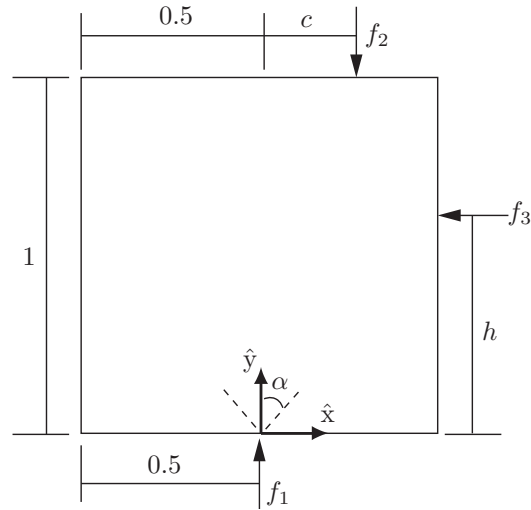


Figure 12.33: A square restrained by three point contacts.

Exercise 12.18 A square is restrained by three point contacts as shown in Figure 12.33: f_1 is a point contact with friction coefficient μ , while f_2 and f_3 are frictionless point contacts. If $c = \frac{1}{4}$ and $h = \frac{1}{2}$, find the range of values of μ such that grasp yields force closure.

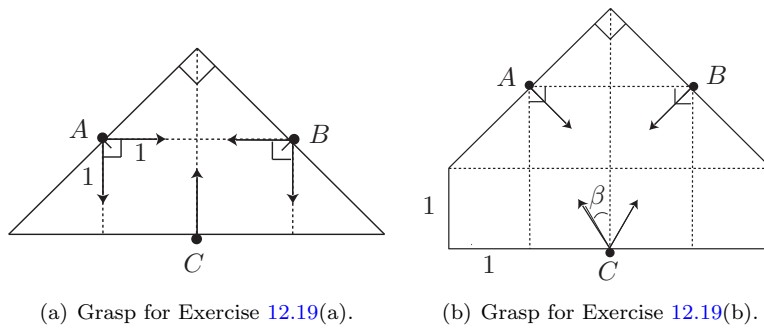


Figure 12.34: Planar grasps.

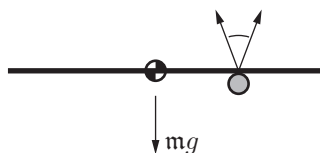


Figure 12.35: A zero-thickness rod supported by a single contact.

Exercise 12.19

- (a) For the planar grasp of Figure 12.34(a), assume contact C is frictionless, while the friction coefficient at contacts A and B is $\mu = 1$. Determine whether this grasp is force closure.
- (b) For the planar grasp of Figure 12.34(b), assume contacts A and B are frictionless, while contact C has a friction cone of half-angle β . Find the range of values of β that makes this grasp force closure.

Exercise 12.20 Find a formula for the minimum friction coefficient, as a function of n , needed for a two-fingered planar force-closure grasp of an n -sided regular polygon, where n is odd. Assume that the fingers can make contact only with the edges, not the vertices. If the fingers could also contact the vertices, how does your answer change? You can assume that the fingers are circular.

Exercise 12.21 Consider a table at rest, supported by four legs in frictional contact with the floor. The normal forces provided by each leg are not unique; there is an infinite set of solutions to the normal forces yielding a force balance with gravity. What is the dimension of the space of normal-force solutions? (Since there are four legs, the space of normal forces is four dimensional, and the space of solutions must be a subspace of this four-dimensional space.) What is the dimension of the space of contact-force solutions if we include tangential frictional forces?

Exercise 12.22 A thin rod in gravity is supported from below by a single stationary contact with friction, shown in Figure 12.35. One more frictionless contact can be placed anywhere else on the top or the bottom of the rod. Indicate all the places where this contact can be put so that the gravitational force is balanced. Use moment labeling to justify your answer. Prove the same using algebraic force balance, and comment on how the magnitude of the normal

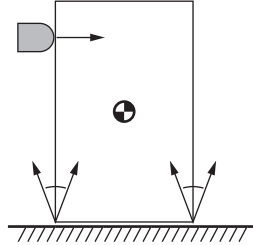


Figure 12.36: A frictionless finger pushes a box to the right. Gravity acts downward. Does the box slide flat against the table, does it tip over the lower right corner, or does it slide and tip over that corner?

forces depends on the location of the second contact.

Exercise 12.23 A frictionless finger begins pushing a box over a table (Figure 12.36). There is friction between the box and the table, as indicated in the figure. There are three possible contact modes between the box and the table: either the box slides to the right flat against the table, or it tips over at the right lower corner, or it tips over that corner while the corner also slides to the right. Which actually occurs? Assume a quasistatic force balance and answer the following questions.

- For each of the three contact modes, draw the moment-labeling regions corresponding to the table's friction cone edges active in that contact mode.
- For each moment-labeling drawing, determine whether the pushing force plus the gravitational force can be quasistatically balanced by the support forces. From this, determine which contact mode actually occurs.
- Graphically show a different support-friction cone for which the contact mode is different from your solution above.

Exercise 12.24 In Figure 12.37 body 1, of mass m_1 with center of mass at (x_1, y_1) , leans on body 2, of mass m_2 with center of mass at (x_2, y_2) . Both are supported by a horizontal plane, and gravity acts downward. The friction coefficient at all four contacts (at $(0, 0)$, at (x_L, y) , at $(x_L, 0)$, and at $(x_R, 0)$) is $\mu > 0$. We want to know whether it is possible for the assembly to stay standing by some choice of contact forces within the friction cones. Write down the six equations of force balance for the two bodies in terms of the gravitational forces and the contact forces, and express the conditions that must be satisfied for this

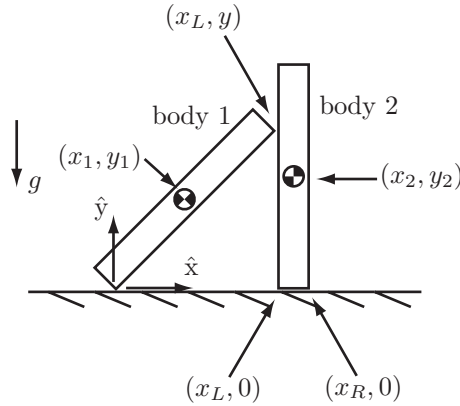


Figure 12.37: One body leaning on another (Exercise 12.24).

assembly to stay standing. How many equations and unknowns are there?

Exercise 12.25 Write a program that accepts a set of contacts acting on a planar body and determines whether the body is in first-order form closure.

Exercise 12.26 Write a program that accepts a set of contacts acting on a spatial body and determines whether the body is in first-order form closure.

Exercise 12.27 Write a program that accepts a friction coefficient and a set of contacts acting on a planar body and determines whether the body is in force closure.

Exercise 12.28 Write a program that accepts a friction coefficient and a set of contacts acting on a spatial body and determines whether the body is in force closure. Use a polyhedral approximation to the friction cone at each contact point that underestimates the friction cone and that has four facets.

Exercise 12.29 Write a program to simulate the quasistatic meter-stick trick of Example 12.10. The program takes as input: the initial x -position of the left finger, the right finger, and the stick's center of mass; the constant speed $\dot{x}$ of the right finger (toward the left finger); and the static and kinetic friction coefficients, where $\mu_s \geq \mu_k$. The program should continue the simulation until

the two fingers touch or until the stick falls. It should plot the position of the left finger (which is constant), the right finger, and the center of mass as a function of time. Include an example where $\mu_s = \mu_k$, an example where μ_s is only slightly larger than μ_k , and an example where μ_s is much larger than μ_k .

Exercise 12.30 Write a program that determines whether a given assembly of planar bodies can remain standing in gravity. Gravity g acts in the $-\hat{y}$ -direction. The assembly is described by m bodies, n contacts, and the friction coefficient μ , all entered by the user. Each of the m bodies is described by its mass m_i and the (x_i, y_i) location of its center of mass. Each contact is described by the index i of each of the two bodies involved in the contact and the unit normal direction (defined as into the first body). If the contact has only one body involved, the second body is assumed to be stationary (e.g., ground). The program should look for a set of coefficients $k_j \geq 0$ multiplying the friction-cone edges at the contacts (if there are n contacts then there are $2n$ friction-cone edges and coefficients) such that each of the m bodies is in force balance, considering gravity. Except in degenerate cases, if there are more force-balance equations ($3m$) than unknowns ($2n$) then there is no solution. In the usual case, where $2n > 3m$, there is a family of solutions, meaning that the force at each contact cannot be known with certainty.

One approach is to have your program generate an appropriate linear program and use the programming language's built-in linear-programming solver.

Exercise 12.31 This is a generalization of the previous exercise. Now, instead of simply deciding whether the assembly stays standing for a stationary base, the base moves according to a trajectory specified by the user, and the program determines whether the assembly can stay together during the trajectory (i.e., whether sticking contact at all contacts allows each body to follow the specified trajectory). The three-dimensional trajectory of the base can be specified as a polynomial in $(x(t), y(t), \theta(t))$, for a base reference frame defined at a particular position. For this problem, you also need to specify the scalar moment of inertia about the center of mass for each body in the assembly. You may find it convenient to express the motion and forces (gravitational, contact, inertial) in the frame of each body and solve for the dynamics in the body frames. Your program should check for stability (all contact normal forces are nonnegative while satisfying the dynamics) at finely spaced discrete points along the trajectory. It should return a binary result: the assembly can be maintained at all points along the trajectory, or not.